

# Final Report for ColorVision App

## 1. Does it accomplish the stated objective? (10 points)

Yes, **ColorVision** successfully accomplishes its stated objective of providing an accessible, affordable, and clinically validated color blindness testing platform.

The app includes two well-known and widely used diagnostic tools for assessing color blindness:

- **Ishihara Test:** This test evaluates red-green color deficiencies, the most common form of color blindness. The test presents users with a series of plates, each containing a pattern of dots that form numbers or paths. Users identify what they see, and their responses are evaluated to determine the presence or absence of color blindness.
- **Farnsworth Test:** This test focuses on a broader range of color vision issues by asking users to arrange color tiles in order of gradient. This test assesses color discrimination ability, providing deeper insights into the type and severity of a user's condition.

In addition to accurate testing, the app includes an **educational section** to raise awareness about color blindness. This section provides:

- **Engaging facts:** Rotating facts about color blindness are displayed in a visually appealing "Fact Box" to help users understand the prevalence, causes, and challenges associated with color vision deficiencies.
- **Resource guidance:** Links and tips for users on how to manage color blindness, including resources like EnChroma glasses and educational content.

Furthermore, the app is designed to be **inclusive and accessible**, making color vision testing available to individuals who may not have access to expensive diagnostic tools or clinical testing facilities. Its affordable, mobile-friendly approach ensures that anyone with a smartphone can test their vision in the comfort of their home.

The combination of these features ensures that ColorVision delivers on its promise of not only testing color vision but also creating awareness and educating users about color blindness. This dual-purpose approach aligns with the app's primary goal of promoting inclusivity and understanding for individuals with color vision deficiencies..

## 2. Does it deliver on the “value proposition”? (10 points)

**ColorVision** delivers on its value proposition by bridging the gap between traditional color blindness testing and modern accessibility needs. It offers affordable, user-friendly tools that can be used anytime and anywhere, making it accessible to a global audience, including those in remote or underserved areas. By providing accurate assessments through the Ishihara and Farnsworth tests, the app ensures that users can gain detailed insights into their color vision without the need for expensive clinical visits. Additionally, the app includes actionable

recommendations, such as using EnChroma glasses or color recognition tools, empowering users to better manage their color vision deficiencies.

It delivers on its value proposition by emphasizing awareness, education, and inclusivity, addressing a critical gap in understanding and managing color vision deficiencies. The app goes beyond simple testing by integrating engaging and educational content that helps users and their communities learn about the challenges of color blindness. Through features such as rotating facts and an educational section, the app fosters greater awareness, making users more informed about their condition and the available tools to support them. This educational component empowers not only those with color vision deficiencies but also educators, parents, and employers, ensuring they can provide meaningful support in academic, professional, and personal environments.

Beyond individual users, the app serves educators, parents, and employers, enabling them to identify and support individuals with color vision deficiencies effectively. With an emphasis on inclusivity, ColorVision promotes awareness through engaging educational content and fun, rotating facts about color blindness. Its intuitive design ensures ease of use for people of all age groups and technical expertise, making it a valuable resource for individuals, families, and organizations alike. By focusing on accessibility, education, and empowerment, the app delivers on its promise to create a more informed and inclusive community.

### **3. Does it respect user privacy/is it secure? (10 points)**

**ColorVision** prioritizes user privacy and incorporates key security measures to protect sensitive information. The app uses local storage to handle critical data, such as test results and user preferences, ensuring that no personal information is transmitted to external servers. This localized approach minimizes exposure to potential breaches and emphasizes privacy-first principles. Additionally, the app collects only the essential information required for functionality, avoiding invasive data practices and ensuring a safe experience for users across all age groups.

To strengthen data protection, ColorVision implements encryption during the sign-up process, safeguarding user credentials against unauthorized access. This ensures secure account creation and login experiences, reducing vulnerabilities associated with data breaches. Future updates aim to introduce more advanced encryption protocols for stored data and provide secure cloud backup options for users seeking multi-device access. These planned enhancements reflect ColorVision's commitment to maintaining high standards of privacy and security while empowering users with confidence in the safety of their personal information.

### **4. Does it have proper app lifecycle/state management? (10 points)**

ColorVision demonstrates robust app lifecycle and state management practices, ensuring a seamless and consistent user experience. User preferences, such as dark mode and high contrast settings, are persistently stored using AsyncStorage, allowing them to be restored

automatically during subsequent sessions. This ensures users do not need to reconfigure their settings each time they access the app, enhancing usability and convenience, particularly for individuals requiring accessibility features. The app's persistent state management approach aligns with modern best practices for mobile app development, reinforcing its reliability and ease of use.

To manage state and navigation effectively, React Native's lifecycle methods are implemented throughout the app. Lifecycle methods like `useEffect` are employed to load user preferences, monitor changes in user settings, and dynamically update content such as the fact box. These methods enable the app to handle events such as app initialization, backgrounding, and foregrounding, ensuring it remains responsive and consistent. For instance, when the app is minimized or reopened, React Native's lifecycle management ensures that the app resumes precisely where the user left off without losing any data or progress, maintaining a smooth and uninterrupted experience.

### **5. Does it address at least three additional challenges (other than privacy and state management) unique to mobile app development? (15 points)**

**ColorVision** addresses three critical challenges unique to mobile app development: device compatibility, offline functionality, and resource constraints. These challenges are tackled with thoughtful design and technical implementation to ensure the app remains accessible, reliable, and high-performing across diverse use cases and user bases.

#### **Device Compatibility**

- The app is designed with responsive layouts and flexible UI components to support various screen sizes and resolutions. Using **React Native's Flexbox-based layout system**, the app ensures a consistent user experience across devices, whether it is a compact smartphone or a larger tablet. This adaptability is further reinforced by conditionally rendering UI elements based on screen dimensions, ensuring that all features, including tests and fact displays, remain visually appealing and functional. Compatibility testing across multiple platforms, such as iOS and Android, further guarantees seamless performance regardless of the operating system, addressing the fragmentation challenge common in mobile app development.

#### **Offline Functionality**

- ColorVision includes offline capabilities to cater to users with limited or unreliable internet access. The app allows users to complete vision tests and access educational content without requiring a live internet connection. This is achieved by preloading critical resources such as test images, fact lists, and educational material during the initial app setup. Storing these resources locally ensures uninterrupted functionality, making the app accessible in remote or low-connectivity areas. This offline functionality not only enhances inclusivity but also aligns with the app's mission of providing accessible tools for color vision assessment and awareness globally.

## **Resource Constraints**

- The app is optimized to minimize storage and memory usage while maintaining high performance. Leveraging lightweight libraries and avoiding unnecessary dependencies ensures the app remains compact and efficient. Images and other assets are compressed and cached locally to reduce download sizes and enhance loading times. Additionally, resource-intensive tasks such as test generation and report creation are executed asynchronously, preventing UI lags or crashes. These optimizations make ColorVision suitable for a wide range of devices, including older smartphones with limited processing power or storage space, ensuring it remains usable for a broad audience.

### **6. Does it have a good user experience? (5 points)**

ColorVision offers an exceptional user experience by combining an intuitive interface, thoughtful accessibility features, and an engaging design. The app ensures that users can navigate effortlessly, regardless of their technical expertise. Its clean layout and responsive design make it easy to locate and use core features like vision tests, educational content, and user preferences. Key accessibility options, such as high contrast mode and dark mode, further enhance usability for individuals with varying needs, ensuring inclusivity for users with visual impairments or color vision deficiencies.

The addition of the rotating fact box introduces an engaging and educational component, making the app more than just a testing tool. This feature encourages users to return to the app, fostering a deeper understanding of color blindness and promoting awareness. By prioritizing simplicity, functionality, and user engagement, ColorVision succeeds in creating a seamless and enjoyable experience for its diverse user base.